

## INDEX

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|                                  |                                       |
|----------------------------------|---------------------------------------|
| <b>EXP NO:</b><br><b>1 DATE:</b> | <b>AZURE DEVOPS ENVIRONMENT SETUP</b> |
|----------------------------------|---------------------------------------|

## AIM:-

To set up and access the Azure DevOps environment by creating an organization through the Azure portal.

## INSTALLATION:-

1. Open your web browser and go to the Azure website: <https://azure.microsoft.com/en-us/get-started/azure-portal>. Sign in using your Microsoft account credentials.

If you don't have a Microsoft account, you can create one here: <https://signup.live.com/?lic=1>

Microsoft Azure portal

Build, deploy, and optimize resources centrally for your Azure account from cloud to edge with an intelligent and intuitive portal for fast and efficient operations.

**Get more done in the portal with Azure Copilot**

Use AI-powered insights and guided actions to simplify management and optimize your Azure workloads.

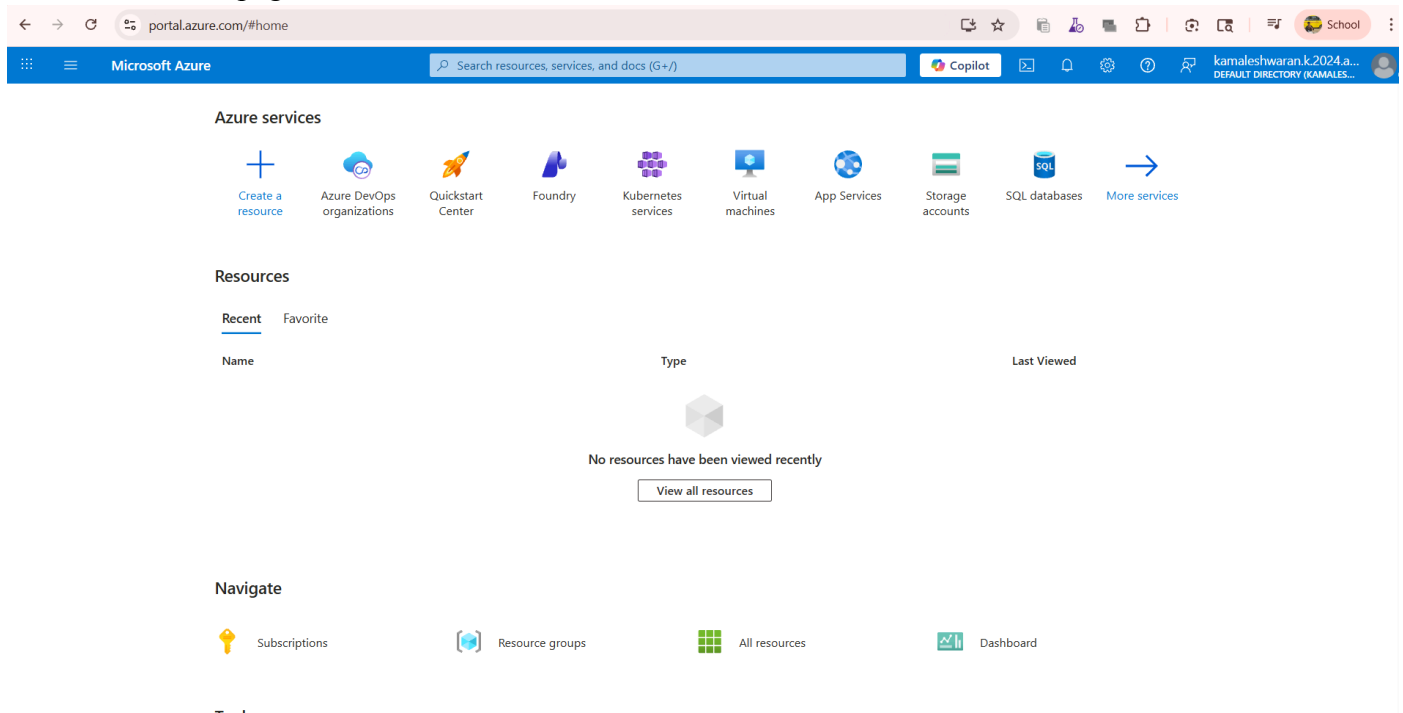
[Sign in now](#) [Get started with Azure](#)

**Resources**

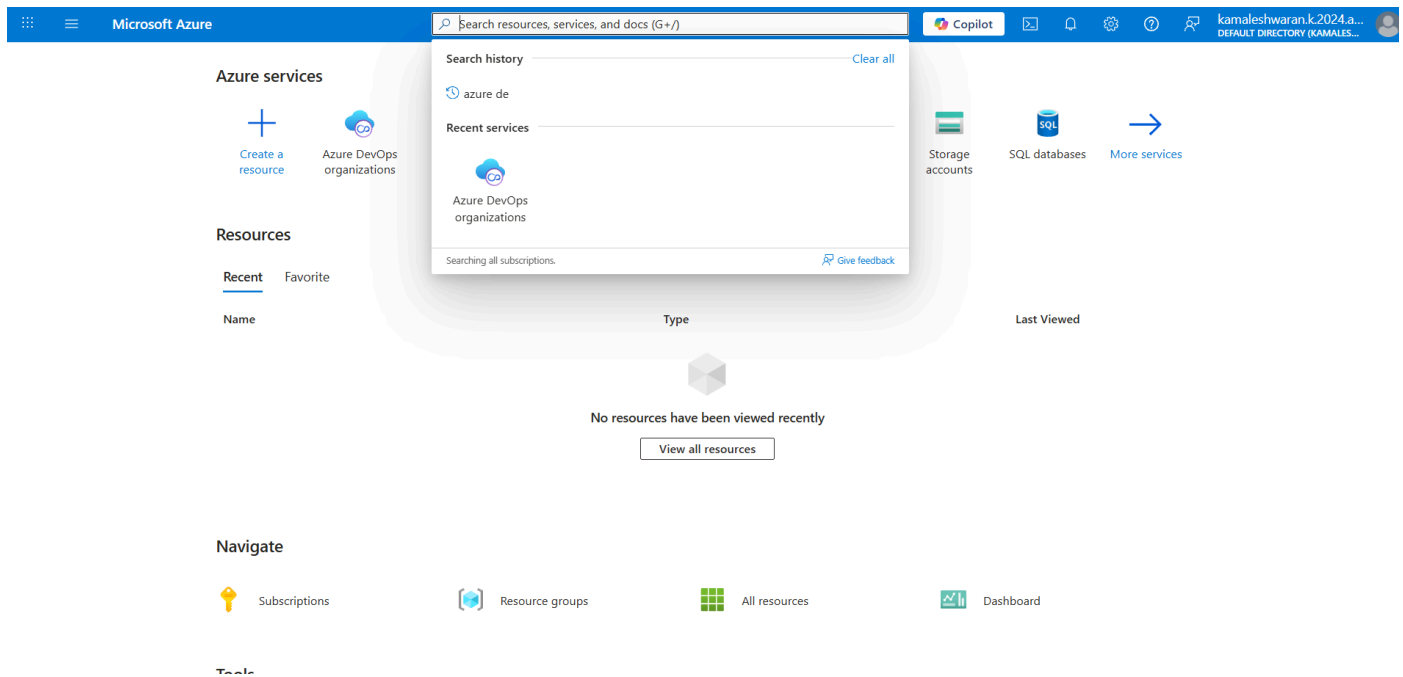
| Resource name | Type   | Alerts | Last checked |
|---------------|--------|--------|--------------|
| East-Asia-2   | Region | 0      | 3 hours ago  |
| East-Asia-2   | Region | 0      | 3 hours ago  |
| East-Asia-2   | Region | 0      | 3 hours ago  |
| East-Asia-2   | Region | 0      | 3 hours ago  |
| East-Asia-2   | Region | 0      | 3 hours ago  |
| East-Asia-2   | Region | 0      | 3 hours ago  |
| East-Asia-2   | Region | 0      | 3 hours ago  |
| East-Asia-2   | Region | 0      | 3 hours ago  |
| East-Asia-2   | Region | 0      | 3 hours ago  |
| East-Asia-2   | Region | 0      | 3 hours ago  |

[Get started with Azure](#)

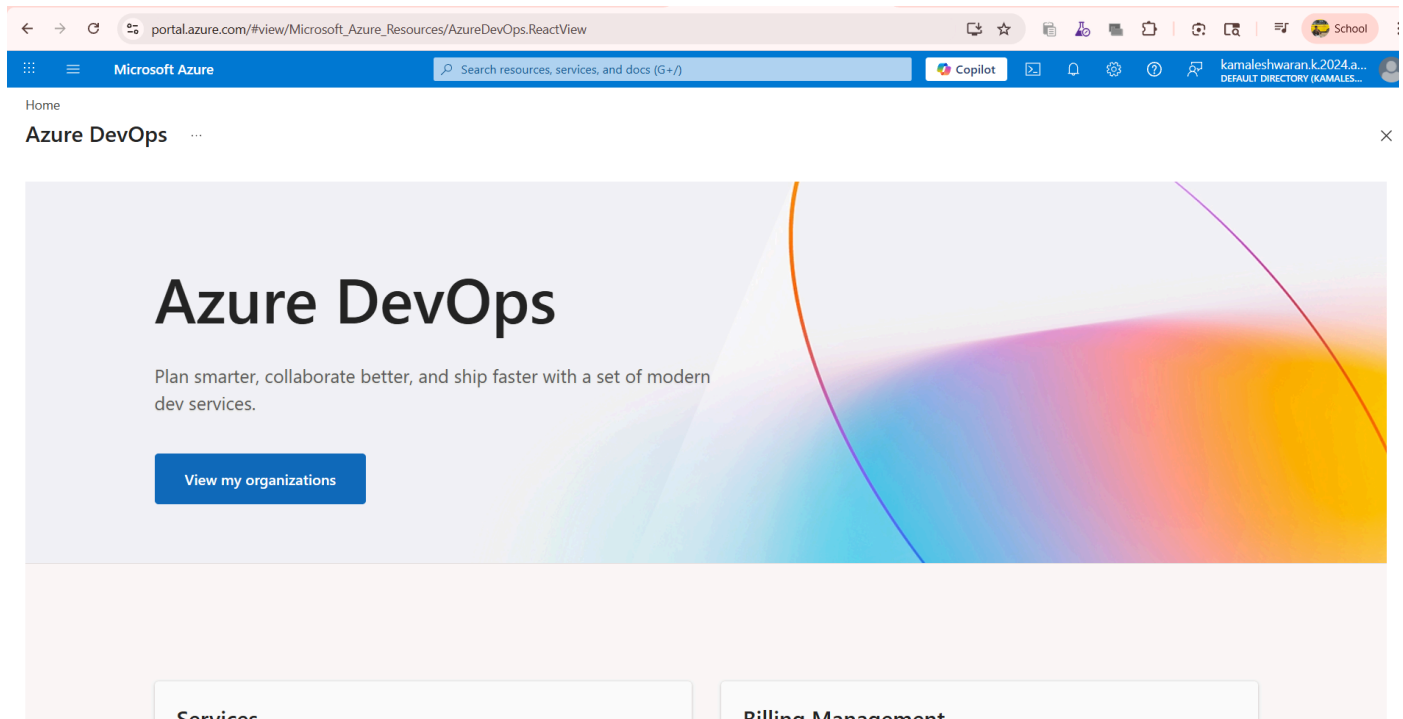
## 2. Azure home page



3. Open DevOps environment in the Azure platform by typing *Azure DevOps Organizations* in the search bar.



4. Click on the ***My Azure DevOps Organization*** link and create an organization and you should be taken to the Azure DevOps Organization Home page.



## **RESULT:**

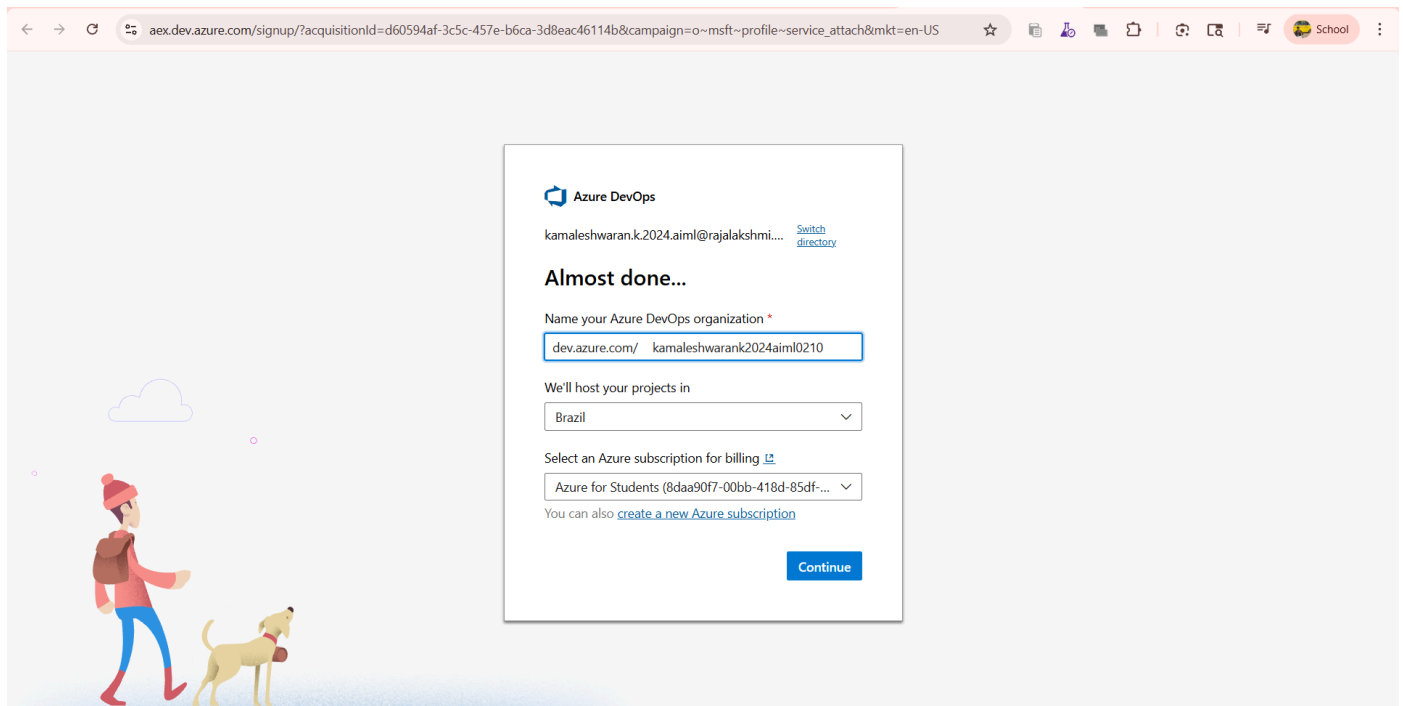
Successfully accessed the Azure DevOps environment and created a new organization through the Azure portal.

|                                  |                                                             |
|----------------------------------|-------------------------------------------------------------|
| <b>EXP NO:</b><br><b>2 DATE:</b> | <b>AZURE DEVOPS PROJECT SETUP AND USER STORY MANAGEMENT</b> |
|----------------------------------|-------------------------------------------------------------|

## **AIM:-**

To set up an Azure DevOps project for efficient collaboration and agile work management.

### 1. Create An Azure Account



aeex.dev.azure.com/signup/?acquisitionId=d60594af-3c5c-457e-b6ca-3d8eac46114b&campaign=o~msft~profile~service\_attach&mkt=en-US

**Azure DevOps**

kamaleshwaran.k.2024.aiml@rajalakshmi... [Switch directory](#)

**Almost done...**

Name your Azure DevOps organization \*

dev.azure.com/ kamaleshwarank2024aiml0210

We'll host your projects in

Brazil

Select an Azure subscription for billing [Link](#)

Azure for Students (8daa90f7-00bb-418d-85df-...)

You can also [create a new Azure subscription](#)

**Continue**

### 2. Create the First Project in Your Organization

- After the organization is set up, you'll need to create your first **project**. This is where you'll begin to manage code, pipelines, work items, and more.
- On the organization's **Home page**, click on the **New Project** button.
- Enter the project name, description, and visibility options:
 

**Name:** Choose a name for the project (e.g., **LMS**).

**Description:** Optionally, add a description to provide more context about the project.

**Visibility:** Choose whether you want the project to be **Private** (accessible only to those invited) or **Public** (accessible to anyone).
- Once you've filled out the details, click **Create** to set up your first project.

## Create new project



Project name \*

Batch Data Analysis and Visualization 1

Description

Visibility



**Private**

Only people you give access to will be able to view this. Want to create a public project? [Try GitHub](#)

^ Advanced

Version control ?

Git



Work item process ?

Agile



Cancel

Create

- Once logged in, ensure you are in the correct organization. If you're part of multiple organizations, you can switch between them from the top left corner (next to your user profile). Click on the Organization name, and you should be taken to the Azure DevOps Organization Home page.



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Microsoft account

India

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---

**Visual Studio Dev Essentials**

Get everything you need to build and deploy your app on any platform.

[Use your benefits](#)

## Azure DevOps Organizations

[Create new organization](#)

- dev.azure.com/kamaleshwarank2024aiml (Owner)
 

**Projects**
  - Online Pharmacy Management System
  - Batch Data Analysis and Visualization
  - Batch Data Analysis and Visualization Team[New project](#)

**Actions**
  - [Open in Visual Studio](#)

---

- > dev.azure.com/mohamedfadiif2024aiml0624 (Member)

#### 4. Project dashboard

dev.azure.com/kamaleshwarank2024aiml/Batch%20Data%20Analysis%20and%20Visualization

Azure DevOps kamaleshwarank2024aiml / Batch Data Analysis and Vis... / Overview / Summary

Search

Private Invite

## BV Batch Data Analysis and Visualization

Batch Data Analysis an... +

Overview

Summary

Dashboards

Wiki

Boards

Repos

Pipelines

Test Plans

Artifacts

### About this project

Help others to get on board!

Describe your project and make it easier for other people to understand it.

+ Add Project Description



### Project stats

Period: Last 7 days

**Boards**

|   |                      |
|---|----------------------|
| 0 | Work items created   |
| 0 | Work items completed |

**Repos**

|   |                      |
|---|----------------------|
| 0 | Pull requests opened |
| 0 | Commits by 0 authors |

**Members** 5



Project settings

5. To manage user stories:

a. From the **left-hand navigation menu**, click on **Boards**. This will take you to the main **Boards**

page, where you can manage work items, backlogs, and sprints.

b. On the **work items** page, you'll see the option to **Add a work item** at the top. Alternatively, you can find a + button or **Add New Work Item** depending on the view you're in. From the **Add a work item** dropdown, select **User Story**. This will open a form to enter details for the new User Story.

The screenshot displays the Azure DevOps interface for a project named "Batch Data Analysis and Visualization Team Team". The view is set to "Backlog". The left-hand navigation menu is visible, with "Backlogs" selected. The main area shows a table of 18 user stories, all in "New" state. The table columns are: Order, Work Item Type, Title, State, Story..., Value Area, Iteration Path, and Tags. The user stories are listed in descending order of ID.

| Order | Work Item Type | Title                                                               | State | Story... | Value Area | Iteration Path                       | Tags |
|-------|----------------|---------------------------------------------------------------------|-------|----------|------------|--------------------------------------|------|
| 1     | User Story     | As a data analyst, I want to upload batch data files (CSV/Exc...    | New   |          | Business   | Batch Data Analysis and Visualiza... |      |
| 2     | User Story     | As a system user, I want validation rules for batch uploads, ...    | New   |          | Business   | Batch Data Analysis and Visualiza... |      |
| 3     | User Story     | As a data engineer, I want scheduled batch ingestion, So th...      | New   |          | Business   | Batch Data Analysis and Visualiza... |      |
| 4     | User Story     | As a data engineer, I want to clean and transform batch dat...      | New   |          | Business   | Batch Data Analysis and Visualiza... |      |
| 5     | User Story     | As a system, I want to store processed batch data securely, ...     | New   |          | Business   | Batch Data Analysis and Visualiza... |      |
| 6     | User Story     | As a system user, I want validation rules for batch uploads, ...    | New   |          | Business   | Batch Data Analysis and Visualiza... |      |
| 7     | User Story     | As a data analyst, I want to perform statistical analysis on b...   | New   |          | Business   | Batch Data Analysis and Visualiza... |      |
| 8     | User Story     | As a data analyst, I want to apply filters and aggregations, S...   | New   |          | Business   | Batch Data Analysis and Visualiza... |      |
| 9     | User Story     | As a system, I want to store processed batch data securely, ...     | New   |          | Business   | Batch Data Analysis and Visualiza... |      |
| 10    | User Story     | As a business user, I want summarized insights from batch a...      | New   |          | Business   | Batch Data Analysis and Visualiza... |      |
| 11    | User Story     | As a system, I want to automatically generate insight report...     | New   |          | Business   | Batch Data Analysis and Visualiza... |      |
| 12    | User Story     | As a data engineer, I want to clean and transform batch dat...      | New   |          | Business   | Batch Data Analysis and Visualiza... |      |
| 13    | User Story     | As a business user, I want interactive dashboards, So that I c...   | New   |          | Business   | Batch Data Analysis and Visualiza... |      |
| 14    | User Story     | As a business user, I want different chart types (bar, pie, line... | New   |          | Business   | Batch Data Analysis and Visualiza... |      |
| 15    | User Story     | As a data analyst, I want to perform statistical analysis on b...   | New   |          | Business   | Batch Data Analysis and Visualiza... |      |
| 16    | User Story     | As a user, I want to export reports in PDF/Excel format, So t...    | New   |          | Business   | Batch Data Analysis and Visualiza... |      |
| 17    | User Story     | As an admin, I want role-based access to reports, So that da...     | New   |          | Business   | Batch Data Analysis and Visualiza... |      |
| 18    | User Story     | As a data analyst, I want to apply filters and aggregations, S...   | New   |          | Business   | Batch Data Analysis and Visualiza... |      |



dev.azure.com/kamaleshwarank2024aiml/Batch%20Data%20Analysis%20and%20Visualization%20Team/\_backlogs/backlog/Batch%20Data%20Analysis%20and%20Visualization%20Team

Azure DevOps kamaleshwarank2024aiml / Batch Data Analysis and Visualization Team / Boards / Backlogs

Batch Data Analysis and Visualization Team Team

Backlog Analytics

|    | Order | Work Item Type | Title                                                                                                      | State | Story... | Value Area | Iteration Path                        | Tags |
|----|-------|----------------|------------------------------------------------------------------------------------------------------------|-------|----------|------------|---------------------------------------|------|
| 1  |       | User Story     | As a data analyst, I want to upload batch data files (CSV/Excel) to the system.                            | New   |          | Business   | Batch Data Analysis and Visualization |      |
| 2  |       | User Story     | As a system user, I want validation rules for batch uploads, so that I can ensure data integrity.          | New   |          | Business   | Batch Data Analysis and Visualization |      |
| 3  |       | User Story     | As a data engineer, I want scheduled batch ingestion, so that I can automate data processing.              | New   |          | Business   | Batch Data Analysis and Visualization |      |
| 4  |       | User Story     | As a data engineer, I want to clean and transform batch data, so that I can analyze it effectively.        | New   |          | Business   | Batch Data Analysis and Visualization |      |
| 5  |       | User Story     | As a system, I want to store processed batch data securely, so that I can ensure data safety.              | New   |          | Business   | Batch Data Analysis and Visualization |      |
| 6  |       | User Story     | As a system user, I want validation rules for batch uploads, so that I can ensure data integrity.          | New   |          | Business   | Batch Data Analysis and Visualization |      |
| 7  |       | User Story     | As a data analyst, I want to perform statistical analysis on batch data, so that I can derive insights.    | New   |          | Business   | Batch Data Analysis and Visualization |      |
| 8  |       | User Story     | As a data analyst, I want to apply filters and aggregations, so that I can refine my analysis.             | New   |          | Business   | Batch Data Analysis and Visualization |      |
| 9  |       | User Story     | As a system, I want to store processed batch data securely, so that I can ensure data safety.              | New   |          | Business   | Batch Data Analysis and Visualization |      |
| 10 |       | User Story     | As a business user, I want summarized insights from batch data, so that I can make informed decisions.     | New   |          | Business   | Batch Data Analysis and Visualization |      |
| 11 |       | User Story     | As a system, I want to automatically generate insight reports, so that I can save time.                    | New   |          | Business   | Batch Data Analysis and Visualization |      |
| 12 |       | User Story     | As a data engineer, I want to clean and transform batch data, so that I can analyze it effectively.        | New   |          | Business   | Batch Data Analysis and Visualization |      |
| 13 |       | User Story     | As a business user, I want interactive dashboards, so that I can visualize data trends.                    | New   |          | Business   | Batch Data Analysis and Visualization |      |
| 14 |       | User Story     | As a business user, I want different chart types (bar, pie, line), so that I can present data effectively. | New   |          | Business   | Batch Data Analysis and Visualization |      |
| 15 |       | User Story     | As a data analyst, I want to perform statistical analysis on batch data, so that I can derive insights.    | New   |          | Business   | Batch Data Analysis and Visualization |      |
| 16 |       | User Story     | As a user, I want to export reports in PDF/Excel format, so that I can share data with others.             | New   |          | Business   | Batch Data Analysis and Visualization |      |
| 17 |       | User Story     | As an admin, I want role-based access to reports, so that I can control data access.                       | New   |          | Business   | Batch Data Analysis and Visualization |      |
| 18 |       | User Story     | As a data analyst, I want to apply filters and aggregations, so that I can refine my analysis.             | New   |          | Business   | Batch Data Analysis and Visualization |      |

Project settings

Microsoft Sign out

KW Kamalesh waran kamaleshwarank.2024aiml@r... My Microsoft account Switch directory

Sign in with a different account

## Result:

setup.

Successfully created an Azure DevOps project with user story management and agile workflow

EXP NO:  
3 DATE:

## SETTING UP EPICS, FEATURES, AND USER STORIES FOR PROJECT PLANNING

### AIM:-

To learn about how to create epics, user story, features, backlogs for your assigned project.

### Create Epic, Features, User Stories, Task

The screenshot shows the Azure DevOps interface for a team named 'Batch Data Analysis and Visualization Team'. The 'Backlog' view is active, displaying a list of work items. The items are organized into three main categories: 1. Data Collection and Batch Processing (Epic), 2. Data Analysis Engine (Epic), and 3. Data Visualization and Reporting (Epic). Each category contains several user stories and features. The interface includes a sidebar with navigation options like Overview, Boards, Work Items, Backlogs, Sprints, Queries, Delivery Plans, Analytics views, Planning Poker, Repos, Pipelines, Test Plans, and Artifacts. The top bar shows the team name and a search bar.

### 1.Fill in Epics

The screenshot shows the details of an Epic titled 'Data Collection and Batch Processing'. The page is divided into several sections: Description, Discussion, Planning, Classification, and Deployment. The 'Description' section is currently empty, with a prompt to 'Click to add Description'. The 'Discussion' section shows a comment box with a placeholder text 'Add a comment. Use # to link a work item, @ to mention a person, or ! to link a pull request.' The 'Planning' section includes fields for Priority, Risk, Effort, Business Value, Time Criticality, Start Date, and Target Date. The 'Classification' section shows the Value area as 'Business'. The 'Deployment' section includes a link to 'Add an existing work item as a parent' and a list of related work items, including 'Batch Data Ingestion' and 'Batch Data Processing'. The page also includes a sidebar with navigation options and a top bar with the team name and search bar.

## 2.Fill in Features

FEATURE 69

Batch Data Ingestion

No one selected

0 Comments

Add Tag

Save and Close

Follow

Updated by Kamallesh waran Apr 19

Statz

New

Area

Batch Data Analysis and Visualization Team

Reason

New

Iteration

Batch Data Analysis and Visualization Team/Sprint 1

Details

4

0

Description

Click to add Description.

Discussion

Add a comment. Use # to link a work item, @ to mention a person, or ! to link a pull request.

Markdown supported. Paste or select files to insert.

Switch to HTML editor

Planning

Priority

2

Risk

Effort

Business Value

Time Criticality

Start Date

Select a date...

Target Date

Select a date...

Classification

Value area

Business

Planning Poker

Loading...

Deployment

To track releases associated with this work item, go to Releases and turn on deployment status reporting for Boards in your pipeline's Options menu. Learn more about deployment status reporting

Development

Add link

Link an Azure Repos commit, pull request or branch to see the status of your development. You can also create a branch to get started.

Related Work

Add link

Parent

67 Data Collection and Batch Processing

Updated Apr 19 @ New

Child

70 As a data analyst, I want to upload batch data files (CSV/Excel)...

Updated 2m ago @ New

72 As a data engineer, I want scheduled batch ingestion. So that ...

Updated 2m ago @ New

71 As a system user, I want validation rules for batch uploads. So t...

Updated 2m ago @ New

## 3.Fill in User Story Details

USER STORY 70

As a data analyst, I want to upload batch data files (CSV/Excel). So that I can process large datasets efficiently.

No one selected

0 Comments

Add Tag

Save and Close

Follow

Updated by Kamallesh waran 3m ago

Statz

New

Area

Batch Data Analysis and Visualization Team

Reason

New

Iteration

Batch Data Analysis and Visualization Team/Sprint 1

Details

1

0

Description

Click to add Acceptance Criteria.

Discussion

Add a comment. Use # to link a work item, @ to mention a person, or ! to link a pull request.

Markdown supported. Paste or select files to insert.

Switch to HTML editor

Planning

Story Points

Priority

2

Risk

Classification

Value area

Business

Planning Poker

Loading...

Deployment

To track releases associated with this work item, go to Releases and turn on deployment status reporting for Boards in your pipeline's Options menu. Learn more about deployment status reporting

Development

Add link

Link an Azure Repos commit, pull request or branch to see the status of your development. You can also create a branch to get started.

Related Work

Add link

Parent

69 Batch Data Ingestion

Updated Apr 19 @ New

**RESULT:-**

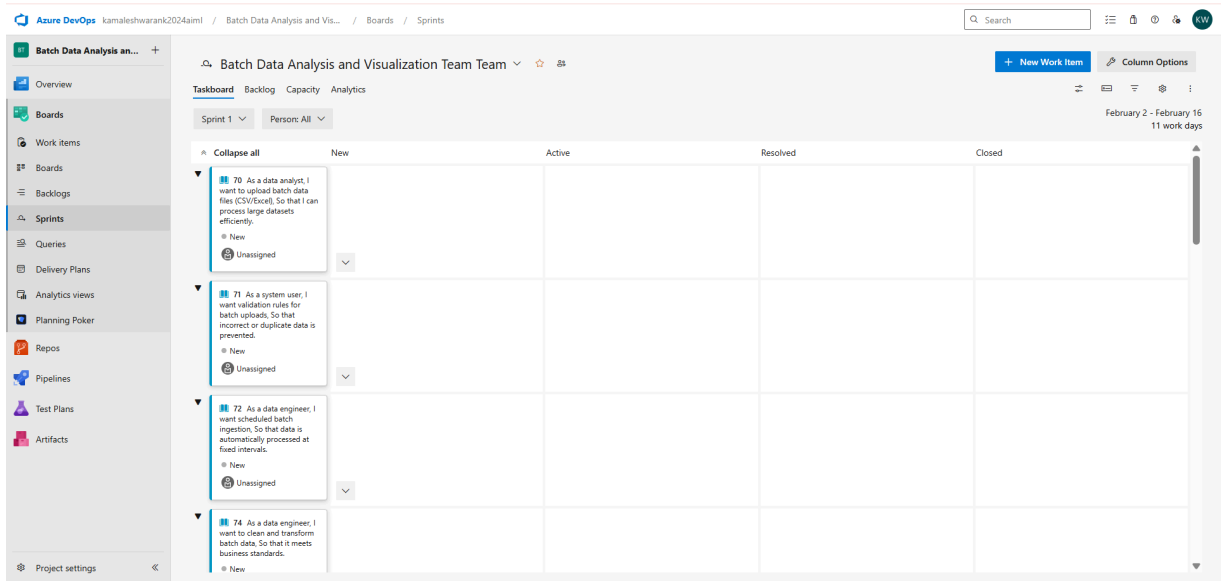
Thus, the creation of epics, features, user story and task has been created successfully.

|                                  |                        |
|----------------------------------|------------------------|
| <b>EXP NO:</b><br><b>4 DATE:</b> | <b>SPRINT PLANNING</b> |
|----------------------------------|------------------------|

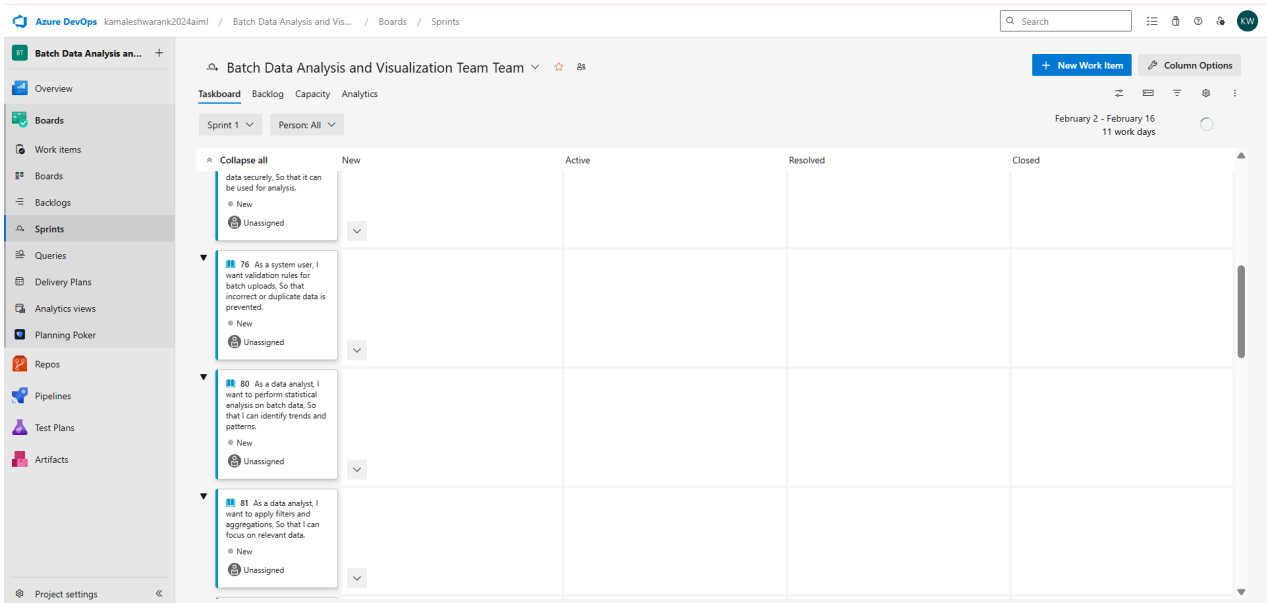
**AIM:-**

To assign user story to specific sprint for the Student Management System Project.

**Sprint Planning:-**  
**Sprint 1**



**Sprint 2**



## Sprint 3

Azure DevOps kamaleshwarank2024aiml / Batch Data Analysis and Vis... / Boards / Sprints

Search

Batch Data Analysis and Visualization Team Team

+ New Work Item Column Options

Taskboard Backlog Capacity Analytics

Sprint 1 Person: All

February 2 - February 16  
11 work days

Collapse all

| New                                                                                                                                                                           | Active | Resolved | Closed |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------|----------|--------|
| <div><div>As a data analyst, I want to perform statistical analysis on batch data, So that I can identify trends and patterns.</div><div>New</div><div>Unassigned</div></div> |        |          |        |
| <div><div>93 As a user, I want to export reports in PDF/Excel format, So that I can share insights easily.</div><div>New</div><div>Unassigned</div></div>                     |        |          |        |
| <div><div>94 As an admin, I want role-based access to reports, So that data security is maintained.</div><div>New</div><div>Unassigned</div></div>                            |        |          |        |
| <div><div>95 As a data analyst, I want to apply filters and aggregations, So that I can focus on relevant data.</div><div>New</div><div>Unassigned</div></div>                |        |          |        |

Project settings

## RESULT:-

The Sprints are created for the Student Management System.

**EXP NO:**  
**5 DATE:**

## POKER ESTIMATION

### AIM:-

Create Poker Estimation for the user stories -Student management System.

### Poker Estimation:-

The screenshot displays the Azure DevOps Planning Poker interface. The top navigation bar shows the URL: dev.azure.com/kamaleshwarank2024aiml/Batch%20Data%20Analysis%20and%20Visualization%20Team/\_apps/hub/SpartezSoftware.plannin... The left sidebar contains a menu with options: Overview, Boards, Work Items, Backlogs, Sprints, Queries, Delivery Plans, Analytics views, Planning Poker (selected), Repos, Pipelines, Test Plans, and Artifacts. The main content area is titled 'Kamalesh Team game' and shows the current item: 'As an admin, I want role-based access to reports. So that data security is maintained.' The interface includes a 'Reveal' button, a row of cards with values 0, 1, 2, 3, 5, 8, and 13, and a 'Reference Items' section. Below this, a table lists work items with columns for Work item, State, and Estimates. The table contains one entry: 'As an admin, I want role-based access to reports. So that data security is maintained.' with a state of 'New' and an estimate of '-'. On the right, a sidebar shows 'USER STORY 94' with the same description, an 'Edit item' button, and a 'Description is empty' message.

### RESULT:-

The Estimation/Story Points is created for the project using Poker Estimation.

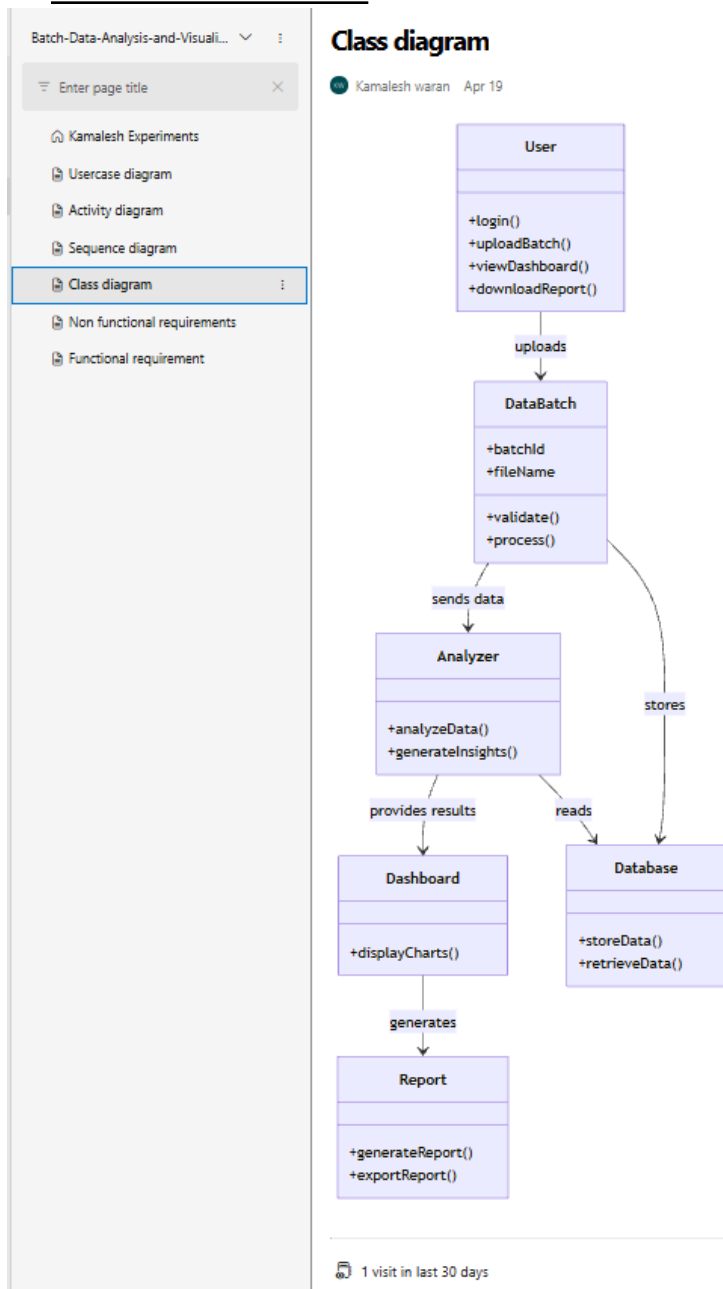
**EXP NO:  
6 DATE:**

## DESIGNING CLASS AND SEQUENCE DIAGRAMS FOR PROJECT ARCHITECTURE

### AIM:-

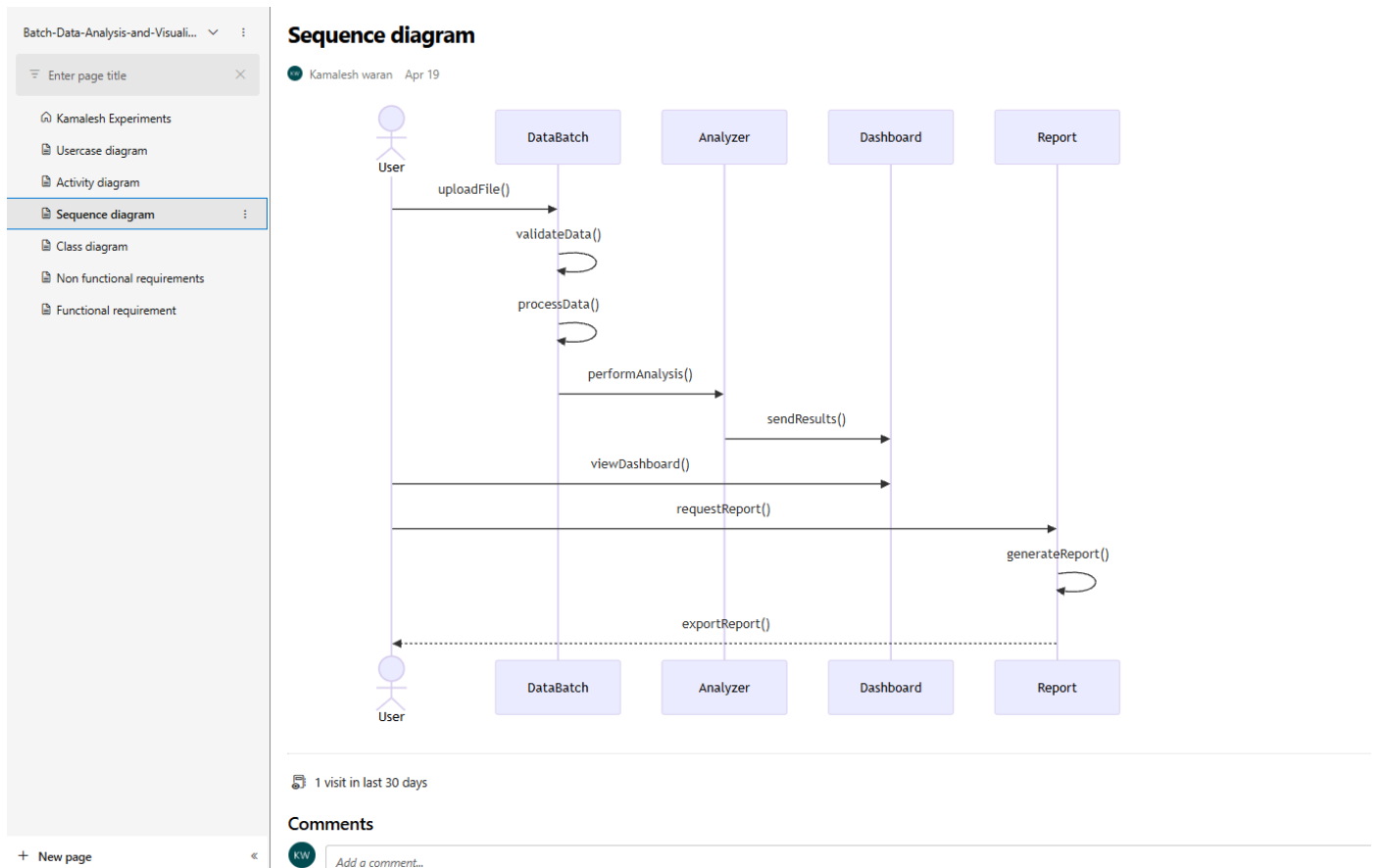
To Design a Class Diagram and Sequence Diagram for the given Project.

### 6A. CLASS DIAGRAM:-





## 6B. SEQUENCE DIAGRAM:-



**Result:**

The Class Diagram and Sequence Diagram is designed Successfully for given project.

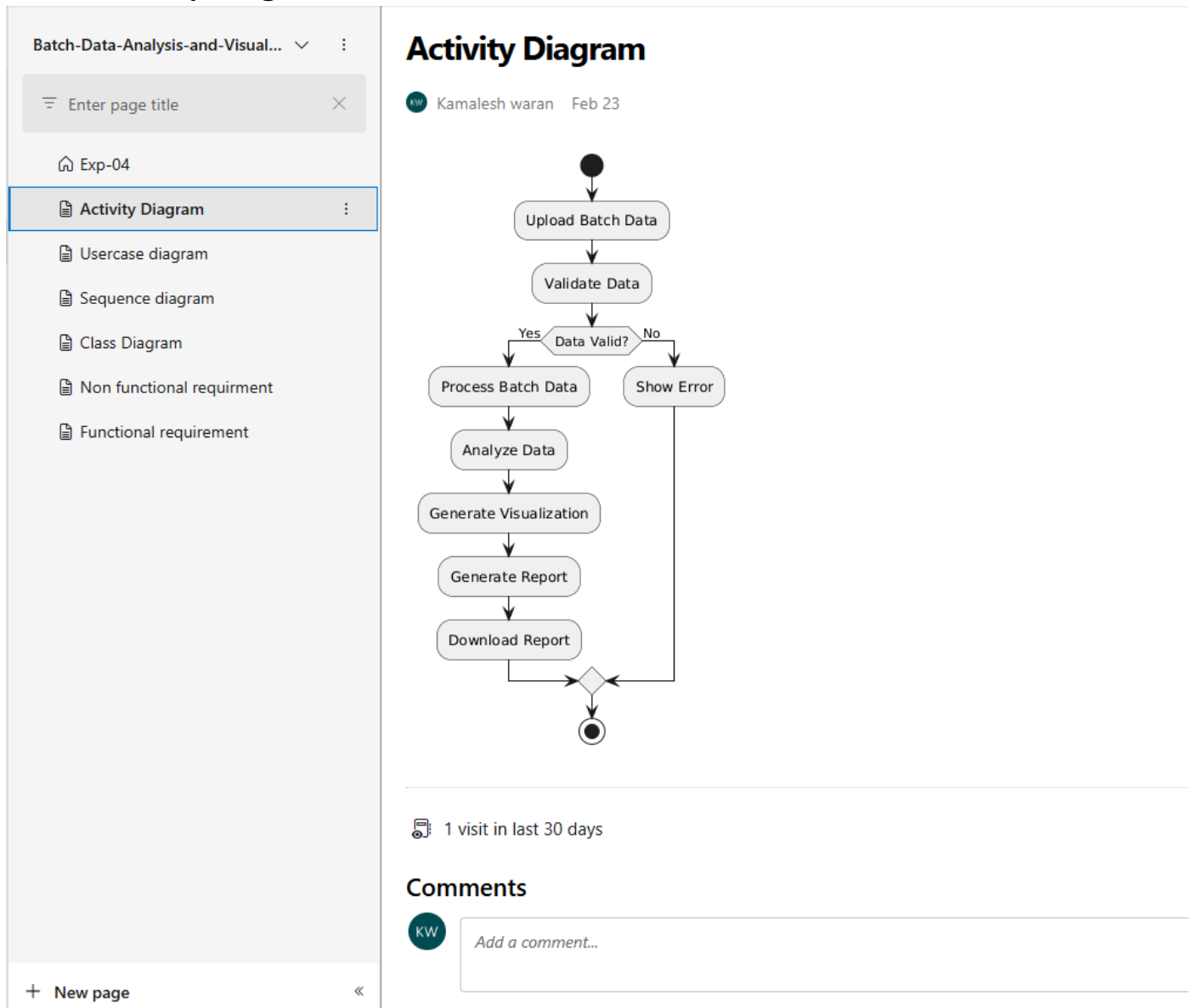
EXP NO:  
7 DATE:

## DESIGNING USE CASE DIAGRAMS AND ACTIVITY DIAGRAM FOR PROJECT ARCHITECTURE

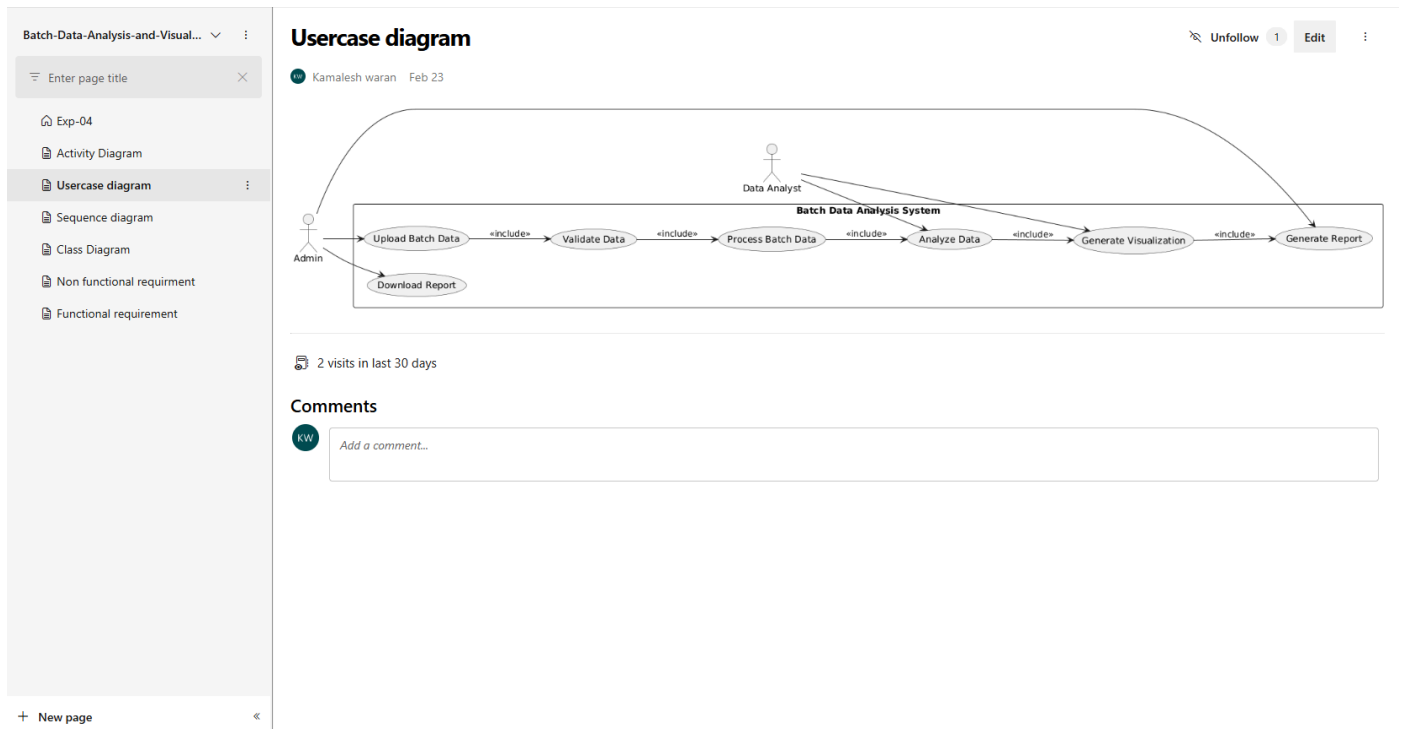
### AIM:-

To Design an activity Diagram and use case Diagram for the given Project.

### 7A. Activity Diagram:-



## 7B. USE CASE DIAGRAM:-



**RESULT:-**

Thus activity and use case diagram has been designed successfully for Student Management System

|                                  |                                            |
|----------------------------------|--------------------------------------------|
| <b>EXP NO:</b><br><b>8 DATE:</b> | <b>TESTING – TEST PLANS AND TEST CASES</b> |
|----------------------------------|--------------------------------------------|

**AIM:-**

Test Plans and Test Case and write two test cases for at least five user stories showcasing the happy path and error scenarios in azure DevOps platform.

**Test Planning and Test Case  
Test Case Design Procedure****1. Understand Core Features of the Application**

- User Signup & Login
- Student Enrollment in Courses
- Assignment Submission & Evaluation
- Viewing Grades and Attendance
- Admin Role Management and Report Generation

**2. Define User Interactions**

- Each test case simulates a real user behavior (e.g., logging in, submitting an assignment, viewing results).

**3. Design Happy Path Test Cases**

- Focused on validating that all core functionalities work correctly under normal conditions.
- **Example:** Student registers and logs in, enrolls in a course, submits assignment, and views grades.

**4. Design Error Path Test Cases**

- Simulate invalid inputs, system issues, or failed actions to ensure proper error handling.
- **Example:** Login with wrong credentials, submission without attachment, unauthorized access to admin panel.

**5. Break Down Steps and Expected Results**

- Each test case includes a clear sequence of actions and expected results.
- Ensures both manual testers and automation tools can follow the process easily.

**6. Use Clear Naming and IDs**

- Test cases are uniquely identified (e.g., TC01 – Student Login Success, TC12 – Invalid Assignment Submission).
- Facilitates easy mapping to features and tracking in Azure DevOps.

**7. Separate Test Suites**

- Grouped by functionality such as:

- Login and Registration
- Course Enrollment

- Assignment Submission
- Report Generation
- Admin Functions
- Improves organization and enables focused execution in Azure DevOps.

## 8. Prioritize and Review

- High-priority assigned to critical workflows like login, course access, and grading.
- Reviewed for completeness, accuracy, and alignment with user stories and feature definitions.

## 1. Test suite

The screenshot shows the Azure DevOps interface for a test suite named 'Batch Data Ingestion (ID: 59)'. The left sidebar contains navigation links: Overview, Boards, Repos, Pipelines, Test Plans, Test plans, Progress report, Runs, Exploratory sessions, and Artifacts. The main area displays the test suite details, including a 'Test Suites' section with a filter by name and a 'Batch Data Ingestion (5)' entry. The 'Test Points (5 items)' table lists the following test cases:

| Test Case Id | Title                              | Outcome | Order | State  | Configuration | Current Tester |
|--------------|------------------------------------|---------|-------|--------|---------------|----------------|
| 61           | TC_01 Login with valid credentials | Passed  | 1     | Design | Windows 10    | Kamalesh waran |
| 62           | TC_02 Login with invalid password  | Passed  | 2     | Design | Windows 10    | Kamalesh waran |
| 63           | TC_03 Login with invalid username  | Passed  | 3     | Design | Windows 10    | Kamalesh waran |
| 64           | TC_04 Login with empty username    | Passed  | 4     | Design | Windows 10    | Kamalesh waran |
| 65           | TC_05 Password Field Masked Check  | Passed  | 5     | Design | Windows 10    | Kamalesh waran |



## 2. Test case

Give two test cases for at least five user stories showcasing the happy path and error scenarios in azure DevOps platform.

Student Management System– Test Plans

### USER STORIES:-

- As a student, I want to sign up and log in securely so I can access academic services. (ID: 101)
- As a student, I need to enroll in courses to participate in classes. (ID: 102)
- As a student, I should be able to submit assignments online for evaluation. (ID: 103)
- As a student, I want to view grades and attendance in one place. (ID: 104)
- As an admin, I want to manage users and generate performance reports. (ID: 105)

### Test Suites:-

**Test suit : TS01 – User Login & Registration (ID: 201)**

#### **1. TC01 – Successful Student Sign-Up**

- **Action:**
  - Go to the Sign-Up page
  - Enter valid name, email, and password
  - Click "Sign Up"
- **Expected Result:**
  - Student account is created and redirected to dashboard
- **Type:** Happy Path

#### **2. TC02 – Login with Valid Credentials**

- **Action:**
  - Go to Login page
  - Enter valid student email & password
  - Click "Login"
- **Expected Result:**
  - Logged in and taken to student dashboard
- **Type:** Happy Path

#### **3. TC03 – Sign-Up with Existing Email**

- **Action:**
  - Go to Sign-Up
  - Enter already registered email
  - Click "Sign Up"
- **Expected Result:**
  - Error message: "Email already registered"
- **Type:** Error Path

#### 4. **TC04 – Login with Incorrect Password**

- **Action:**
  - Enter wrong password
  - Click "Login"
- **Expected Result:**
  - Error: "Invalid username or password"
- **Type:** Error Path

#### **Test Suite: TS02 – Course Enrollment (ID: 202)**

##### 1. **TC05 – Enroll in Available Course**

- **Action:**
  - Login → Navigate to Course List
  - Select a course → Click "Enroll"
- **Expected Result:**
  - Enrollment confirmation shown
- **Type:** Happy Path

##### 2. **TC06 – Try Enrolling in a Full Course**

- **Action:**
  - Select a course with no seats available
  - Click "Enroll"
- **Expected Result:**
  - Error: "Enrollment failed – course full"
- **Type:** Error Path

#### **Test Suite: TS03 – Assignment Submission (ID: 203)**

##### 1. **TC07 – Submit Assignment Successfully**

- **Action:**
  - Login → Go to Assignments
  - Upload file and click "Submit"
- **Expected Result:**
  - Submission confirmation message shown
- **Type:** Happy Path

##### 2. **TC08 – Submit Without Uploading File**

- **Action:**
  - Open assignment submission
  - Click "Submit" without file
- **Expected Result:**
  - Error: "Please upload a file before submitting"
- **Type:** Error Path

## **Test Suite: TS04 – View Grades & Attendance (ID: 204)**

### **1. TC09 – View Grades Dashboard**

- **Action:**
  - Login → Go to "My Grades"
- **Expected Result:**
  - All grades are displayed by subject
- **Type:** Happy Path

### **2. TC10 – Fail to Fetch Grades (Offline Scenario)**

- **Action:**
  - Disconnect network → Go to "My Grades"
- **Expected Result:**
  - Error: "Unable to fetch data – check connection"
- **Type:** Error Path

## **Test Suite: TS05 – Admin Management (ID: 205)**

### **1. TC11 – Assign Role to New User**

- **Action:**
  - Login as Admin → Navigate to User Management
  - Select user and assign role
- **Expected Result:**
  - Confirmation: "Role assigned successfully"
- **Type:** Happy Path

### **2. TC12 – Generate Report with No Data**

- **Action:**
  - Navigate to Reports → Choose empty semester
  - Click "Generate"
- **Expected Result:**
  - Message: "No data available to generate report"
- **Type:** Error Path

### 3. Test Cases

Batch Data Analysis an...

+

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Batch Data Ingestion (ID: 59)

Feb 2 - Feb 16

Past

100% run, 100% passed.

View report

Test Suites

Filter suites by name

Batch Data Ingestion (5)

Batch Data Ingestion (ID: 59)

Execute

Chart

TEST CASE 61

61 TC0\_01 Login with valid credentials

Kamalesh waran

0 Comments

Add Tag

Save

State

Design

Reason

New

Area

Batch Data Analysis and Visualization

Iteration

Batch Data Analysis and Visualization/Sprint 1

Updated by Kamalesh waran: Apr 6

Steps

Summary

Associated Automation

Steps

1. Open login page

User is redirected to dashboard

2. Enter valid username

move to password

3. Enter valid password

verify the password

4. Click Login+

open website

Click or type here to add a step

Recent test results

Passed by with Windows 10

Completed 8:36 AM, 4/19/2026

Failed by with Windows 10

Completed 8:36 AM, 4/19/2026

Passed by with Windows 10

Completed 8:36 AM, 4/19/2026

Passed by with Windows 10

Completed 8:36 AM, 4/19/2026

View all test result history

Deployment

To track releases associated with this work item, go to Releases and turn on deployment status reporting for Boards in your pipeline's Options menu. Learn more about deployment status reporting

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Past

100% run, 100% passed.

View report

Test Suites

Filter suites by name

Batch Data Ingestion (5)

Batch Data Ingestion (ID: 59)

Execute

Chart

Search Title, Test case id and Tags columns

Outcome

Current Tester

Configuration

Assigned To

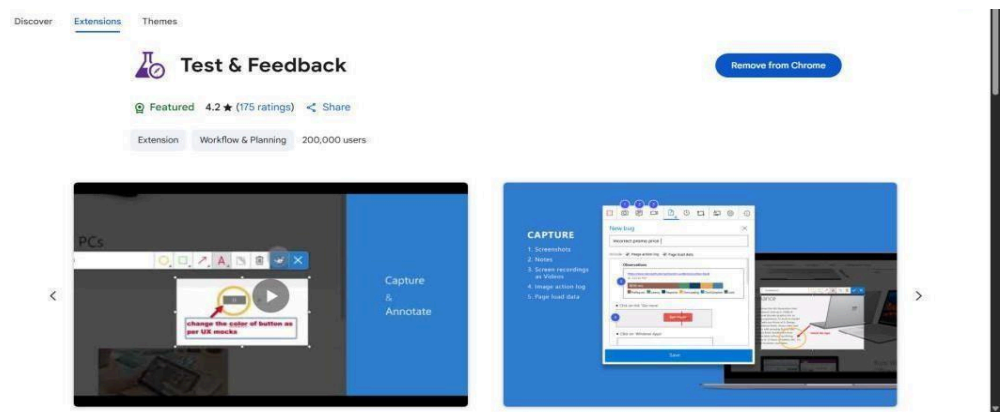
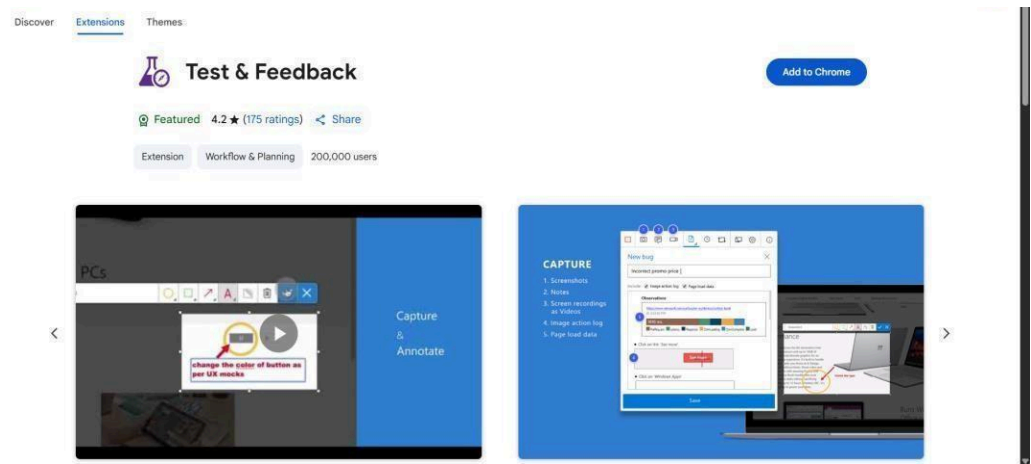
State

Test Points (5 items)

| Title                                                                   | Outcome | Order | Test Case Id | State  | Configuration | Current Tester |
|-------------------------------------------------------------------------|---------|-------|--------------|--------|---------------|----------------|
| <input checked="" type="checkbox"/> TC0_01 Login with valid credentials | Passed  | 1     | 61           | Design | Windows 10    | Kamalesh waran |
| <input type="checkbox"/> TC_02 Login with invalid password              | Passed  | 2     | 62           | Design | Windows 10    | Kamalesh waran |
| <input type="checkbox"/> TC_03 Login with invalid username              | Passed  | 3     | 63           | Design | Windows 10    | Kamalesh waran |
| <input type="checkbox"/> TC_04 Login with empty username                | Passed  | 4     | 64           | Design | Windows 10    | Kamalesh waran |
| <input type="checkbox"/> TC_05 Password Field Masked Check              | Passed  | 5     | 65           | Design | Windows 10    | Kamalesh waran |



4. Installation of test



# 5. Running the test cases

Azure DevOps

kamaleshwarank2024aiml / Batch Data Analysis and Vis... / Test Plans / Batch Data Ingestion

Batch Data Analysis an...

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Batch Data Ingest...

Feb 2 - Feb 16 Past

100% run, 100% passed. [View report](#)

Test Suites

Filter suites by name

Batch Data Ingestion (5)

Batch Data Ingestion (ID: 59)

Execute Chart

Test Points (5 items)

| Title                                                                   | Outcome | Order | Test Case Id | State  | Configuration | Current Tester |
|-------------------------------------------------------------------------|---------|-------|--------------|--------|---------------|----------------|
| <input checked="" type="checkbox"/> TC0_01 Login with valid credentials | Passed  | 1     | 61           | Design | Windows 10    | Kamalesh waran |
| <input type="checkbox"/> TC_02 Login with invalid password              | Passed  | 2     | 62           | Design | Windows 10    | Kamalesh waran |
| <input type="checkbox"/> TC_03 Login with invalid username              | Passed  | 3     | 63           | Design | Windows 10    | Kamalesh waran |
| <input type="checkbox"/> TC_04 Login with empty username                |         |       | 64           | Design | Windows 10    | Kamalesh waran |
| <input type="checkbox"/> TC_05 Password Field Masked Check              |         |       | 65           | Design | Windows 10    | Kamalesh waran |

View execution history

Mark Outcome

Run

Reset test to active

Edit test case

View test result

Run for web application

Run for desktop application

Run with options

Save and close

Create bug

44\*: Verify whether you can add new users to the application.

1. open the website in the browser

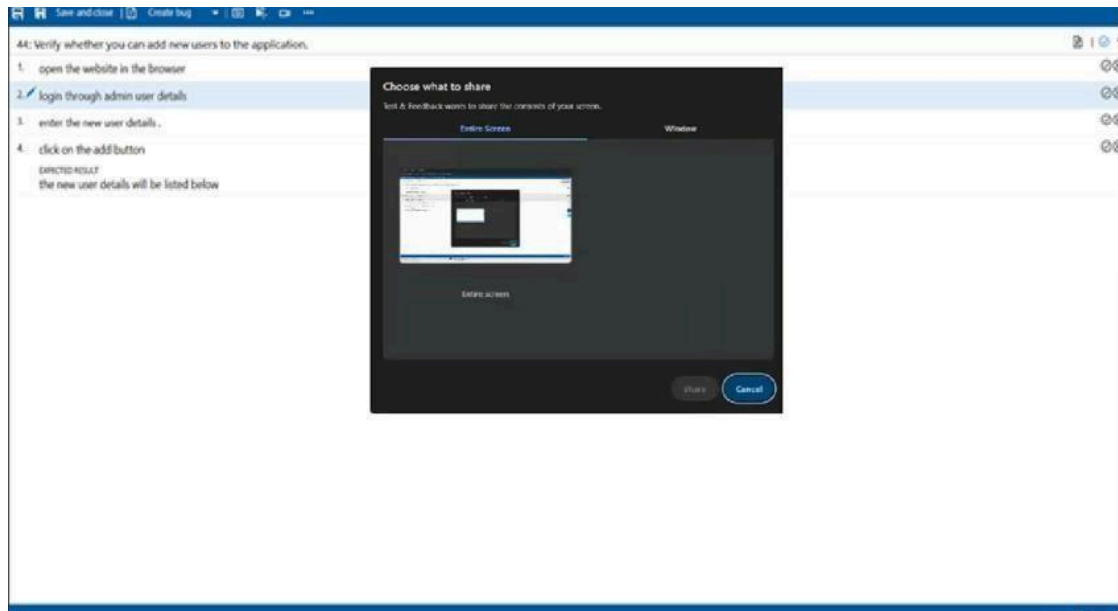
2. login through admin user details

3. enter the new user details .

4. click on the add button

EXPECTED RESULT  
the new user details will be listed below

## 6. Recording the test case



## 7. Creating the bug

Batch Data Analysis an... +

Recently updated Back to Work Items 5 of 34

BUG 39 Invalid Credentials

No one selected 0 Comments Add Tag

Save Follow

Updated by Prasanth H: Monday

Details

State: New Area: Batch Data Analysis and Visualization Reason: New Iteration: Batch Data Analysis and Visualization\Sprint 1

**System Info**

|                                         |                                                                                                                 |
|-----------------------------------------|-----------------------------------------------------------------------------------------------------------------|
| Browser - Name                          | Google Chrome 146                                                                                               |
| Browser - Language                      | en-GB                                                                                                           |
| Browser - Height                        | 1032                                                                                                            |
| Browser - Width                         | 1920                                                                                                            |
| Browser - User agent                    | Mozilla/5.0 (Windows NT 10.0; Win64; x64) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/146.0.0.0 Safari/537.36 |
| Operating system - Name                 | Windows NT 10.0; Win64; x64                                                                                     |
| Operating system - Architecture         | x86_64                                                                                                          |
| Operating system - Processor model      | 13th Gen Intel(R) Core(TM) i9-13900                                                                             |
| Operating system - Number of processors | 32                                                                                                              |
| Memory - Available                      | 22793551872                                                                                                     |
| Memory - Capacity                       | 34033348608                                                                                                     |
| Display - Pixels per inch (X axis)      | 96                                                                                                              |
| Display - Pixels per inch (Y axis)      | 96                                                                                                              |
| Display - Diagonal resolution           | 1                                                                                                               |

Resolved Reason

Story Points

Priority: 2

Severity: 3 - Medium

Activity

**Effort (Hours)**

Original Estimate

Remaining

To track releases associated with this work item, go to [Releases](#) and turn on deployment status reporting for Boards in your pipeline's Options menu. [Learn more about deployment status reporting](#)

**Development**

Add link

Link an Azure Repos [commit](#), [pull request](#) or [branch](#) to see the status of your development. You can also [create a branch](#) to get started.

**Related Work**



## Test Results

Azure DevOps

kamaleshwaran2024aiml / Batch Data Analysis and Vis... / Test Plans / Batch Data Ingestion

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Batch Data Ingestio...

Feb 2 - Feb 16 Past

100% run, 100% passed. View report

Test Suites

Filter suites by name

Batch Data Ingestion (5)

Batch Data Ingestion (ID: 59)

Execute Chart

Search Title, Test case id and Tags columns

Outcome Current Tester Configuration Assigned To State

Test Points (5 items) Run for web application

| Title                                                                  | Outcome | Order | Test Case Id | State  | Configuration | Current Tester |
|------------------------------------------------------------------------|---------|-------|--------------|--------|---------------|----------------|
| <input checked="" type="checkbox"/> TC_01 Login with valid credentials | Passed  | 1     | 61           | Design | Windows 10    | Kamalesh waran |
| <input type="checkbox"/> TC_02 Login with invalid password             | Passed  | 2     | 62           | Design | Windows 10    | Kamalesh waran |
| <input type="checkbox"/> TC_03 Login with invalid username             | Passed  | 3     | 63           | Design | Windows 10    | Kamalesh waran |
| <input type="checkbox"/> TC_04 Login with empty username               | Passed  | 4     | 64           | Design | Windows 10    | Kamalesh waran |
| <input type="checkbox"/> TC_05 Password Field Masked Check             | Passed  | 5     | 65           | Design | Windows 10    | Kamalesh waran |

8. Test report summary

Batch Data Analysis an...

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Batch Data Ingestion (5)

Batch Data Ingestion (ID: 59)

Execute

Test Points (5 items)

TC\_01 Login with valid credentials

TC\_02 Login with invalid password

TC\_03 Login with invalid username

TC\_04 Login with empty username

TC\_05 Password Field Masked Check

Passed

Passed

Passed

Passed

Passed

Test point ID 2 execution history

Test Case61 TC0\_01 Login with valid credentials

Test Suite59 Batch Data Ingestion

Current TesterKamalesh waran

ConfigurationWindows 10

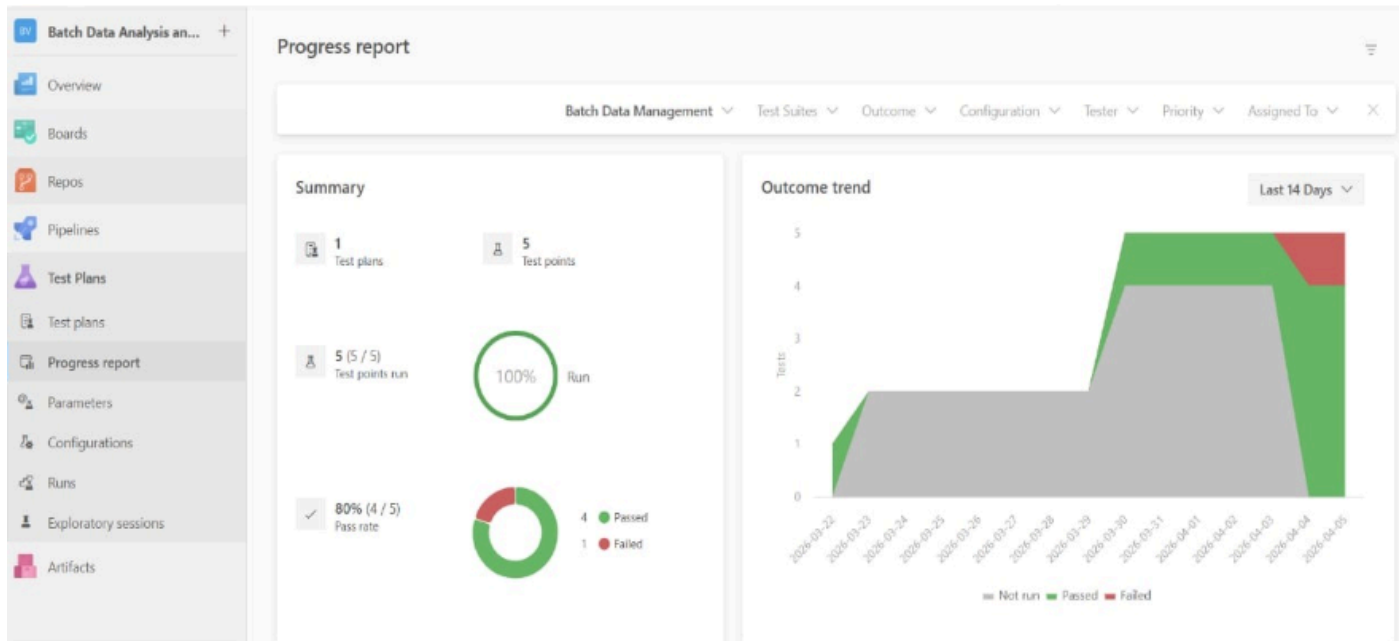
| Outcome        | Test run | Run by         | Time completed |
|----------------|----------|----------------|----------------|
| Passed         | 17       | Kamalesh waran | Apr 19         |
| Failed         | 16       | Kamalesh waran | Apr 19         |
| Passed         | 14       | Kamalesh waran | Apr 19         |
| Passed         | 13       | Kamalesh waran | Apr 19         |
| Not Applicable | 12       | Kamalesh waran | Apr 19         |
| Passed         | 4        | Kamalesh waran | Apr 6          |

Currently showing a test point execution history. View all history for a full test case execution history.

View all history

1. Assigning bug to the developer and changing state

## 9. Progress report



## 10.Changing the test template

Azure DevOps

kamaleshwarank2024aiml / Settings / Process

Q Search

KW

Organization Setti...

kamaleshwarank2024aiml

Q Search Settings

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Processes

Fields

Help

Filter by process name

| Name            | Description                                                                                                                   | Team projects |
|-----------------|-------------------------------------------------------------------------------------------------------------------------------|---------------|
| Basic (default) | This template is flexible for any process and great for teams getting started with Azure DevOps.                              | 1             |
| Agile           | This template is flexible and will work great for most teams using Agile planning methods, including those practicing Scrum.  | 2             |
| Scrum           | This template is for teams who follow the Scrum framework.                                                                    | 0             |
| CMMI            | This template is for more formal projects requiring a framework for process improvement and an auditable record of decisions. | 0             |

**Result:**

The test plans and test cases for the user stories is created in Azure DevOps with Happy Path and Error Path

|                                  |                                 |
|----------------------------------|---------------------------------|
| <b>EXP NO:</b><br><b>9 DATE:</b> | <b>CI/CD PIPELINES IN AZURE</b> |
|----------------------------------|---------------------------------|

**AIM:**

To implement a Continuous Integration and Continuous Deployment (CI/CD) pipeline in Azure DevOps for automating the build, testing, and deployment process of the Student Management System, ensuring faster delivery and improved software quality.

**PROCEDURE:****Steps to Create and implement pipelines in Azure:**

1. Sign in to Azure DevOps and Navigate to Your Project  
Log in to [dev.azure.com](https://dev.azure.com), select your organization, and open the project where your Student Management System code resides.
2. Connect a Code Repository (Azure Repos or GitHub)  
Ensure your application code is stored in a Git-based repository such as Azure Repos or GitHub. This will be the source for triggering builds and deployments in your pipeline.
3. Create a New Pipeline  
Go to the Pipelines section on the left panel and click “Create Pipeline”.  
Choose your source (e.g., Azure Repos Git or GitHub), and then select the repository containing your project code.
4. Choose the Pipeline Configuration  
You can select either the YAML-based pipeline (recommended for version control and automation) or the Classic Editor for a GUI-based setup.  
If using YAML, Azure DevOps will suggest a template or allow you to define your own.
5. Define Build Stage (CI - Continuous Integration) from YAML file
6. Install dependencies (e.g., npm install, dotnet restore)
7. Build the application (dotnet build, npm run build)
8. Run unit tests (dotnet test, npm test)
9. Publish build artifacts to be used in the release stage

10. Save and Run the Pipeline for the First Time

Save the YAML or build definition and click “Run”.

Azure will fetch the latest code and execute the defined build and test stages.

11. Configure Continuous Deployment (CD)

Navigate to the Releases tab under Pipelines and click “New Release Pipeline”.

Add an Artifact (from the build stage) and create a new Stage (e.g., Development, Production).

12. Configure the CD stage with deployment tasks such as deploying to Azure App Service, running database migrations or scripts, and restarting services using the Azure App Service Deploy task linked to your subscription and app details.

13. Set Triggers and Approvals

Enable continuous deployment trigger so the release pipeline runs automatically after a successful build.

For production environments, configure pre-deployment approvals to ensure manual verification before release.

14. Monitor Pipelines and Manage Logs

View all pipeline runs under the Runs section.

Check logs for build/test/deploy stages to debug any errors.

You can also integrate email alerts or Microsoft Teams notifications for build failures.

15. Review and Maintain Pipelines

Regularly update your pipeline tasks or YAML configurations as your application grows.

Ensure pipeline runs are clean and artifacts are stored securely.

Integrate quality gates and code coverage policies to maintain code quality.

dev.azure.com/kamleshwarank2024aiml/Batch%20Data%20Analysis%20and%20Visualization/\_build/results?buildId=1&view=results

Search

School

Azure DevOps

kamleshwarank2024aiml / Batch Data Analysis and Vis... / Pipelines / KamleshwarankSoftware-... / 20260504.1

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#20260504.1 • Set up CI with Azure Pipelines

Run new

This run is being retained as one of 3 recent runs by pipeline.

View retention leases

Summary

Code Coverage

Individual CI by Kamleshwarank

Repository and version

Time started and elapsed

Related

Tests and coverage

View change

Parameters

Stages

Jobs

Stage

1 job completed

7s

Job

3s

Rerun Stage





**RESULT:**

Thus the pipelines for the given project “Student Management System has been executed successfully

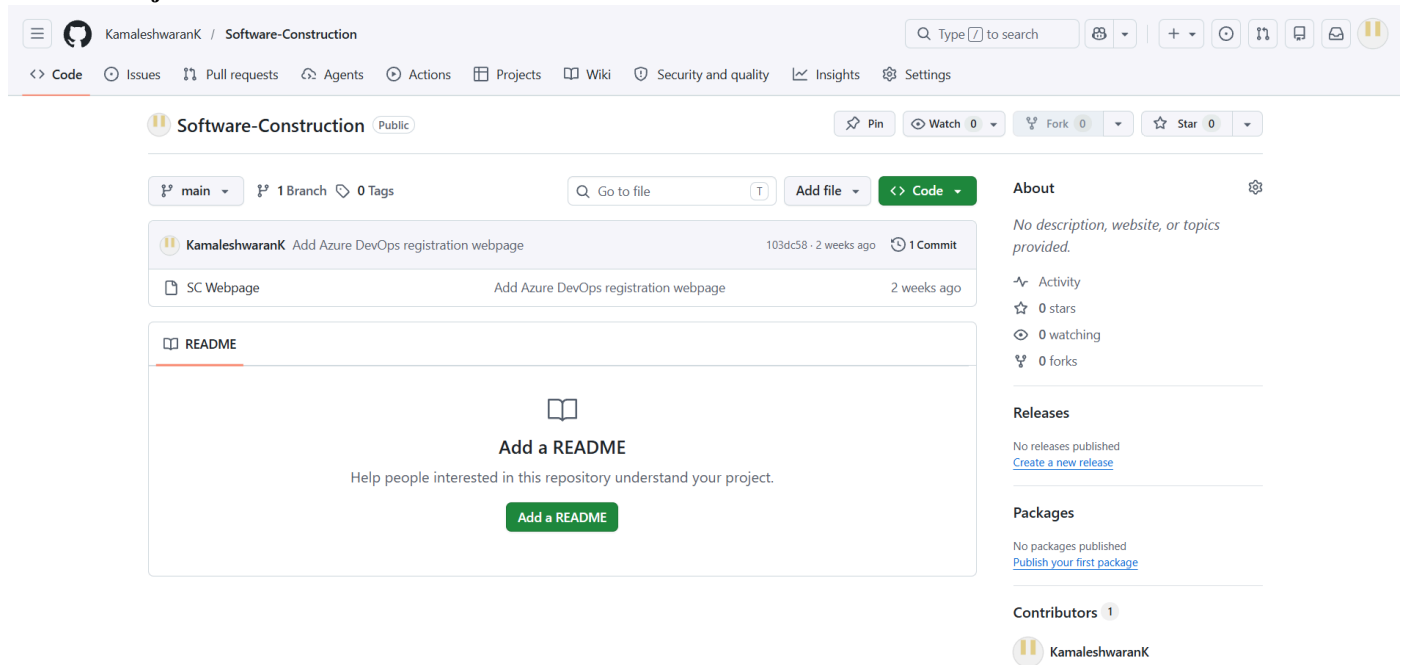
**EXP NO: 10**  
**DATE:**

## **GITHUB: PROJECT STRUCTURE & NAMING CONVENTIONS**

### **Aim:**

To provide a clear and organized view of the project's folder structure and file naming conventions, helping contributors and users easily understand, navigate, and extend the Student Management System project.

### **GitHub Project Structure**



### **Result:**

The GitHub repository clearly displays the organized project structure and consistent naming

conventions, making it easy for users and contributors to understand and navigate the codebase.